

Assignment 6

Due Sunday October 26th, 2025, Self Grading Due Sunday November 2nd, 2025

1. The pulse shape $p(t)$ of a baseband binary PAM system is defined by

$$p(t) = \text{sinc}\left(\frac{t}{T_b}\right)$$

where T_b is the bit duration of the input binary data. The amplitude levels at the pulse generator are +1 or -1, depending on whether the binary symbol at the input is 1 or 0, respectively. Describe the waveform in response to the input data sequence 001101001. Do the symbols interfere with each other?

2. Madhow Problem 4.1
3. Madhow Problem 4.5
4. Madhow Problem 4.11a-c
5. **Self-grade Homework 4.** Fill out the self-grade rubric for Homework 4. Attach to Homework 6 submission.